

Angelina Zhou

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EDUCATION

Cornell University

Ithaca, NY

Computer Science Major | GPA: 4.0

Aug. 2023 – May 2027

Relevant Coursework: Algorithms, OOP, Data Structures, Functional Programming, Data Science, Machine Learning

Awards: Gates Scholarship Cohort VI (one of 750 selected from 51,000+ applicants), IEEE Executive Board

WORK EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Google

Kirkland, WA

- Incoming Intern on Google Cloud BigQuery team, building chat history management tooling for LLM agents

Software Engineering Intern

June 2025 – Aug. 2025

Capital One

- Built an internal rules authoring UI with TypeScript and Vue to generate JSON for credit card decision workflows
- Reduced authoring errors and increased entry speed by 33%, improving efficiency of credit card approval processes
- Implemented Pinia data stores and data models for rules to be passed into backend APIs and stored in DynamoDB
- Completed Capital One's full-stack development curriculum with React, Node.js, FastAPI, PostgreSQL, and AWS

RESEARCH & ACADEMIC EXPERIENCE

Research Assistant

Jan. 2025 – Present

Cornell Napp Lab

- Improved bee color tag detection with OpenCV using RGB to HSV conversion, thresholding, morphological erosion
- Distinguished yellow bee tags from yellow pollen by detecting black tag numbers, reducing false positive rates from 30.6% to 16.7% and 40.4% to 8.6% in test trials
- Built a Flask app to verify tag color with one click, automating file retrieval and replacing manual drag-and-drop

Teaching Assistant, Introduction to Analysis of Algorithms

Aug. 2025 – Dec. 2025

Cornell University

- Supported 250+ students in office hours, teaching graph algorithms (Dijkstra's, Bellman-Ford), binary search, dynamic programming, max flow, FFT, and complexity theory (P vs NP)

Research Assistant

Jan. 2025 – May 2025

Cornell Zhang Research Group

- Designed and implemented a team of LLM agents using Python and AutoGen to automate and accelerate Verilog hardware design
- Developed an iterative feedback-driven workflow where LLM agents generate Verilog code, execute test benches, analyze results, and improve designs autonomously

PROJECTS

Etch-A-Sketch | JavaScript, HTML, CSS, OpenAI API, Vercel, [GitHub](#)

Aug. 2024 – Jan. 2026

- Built a full-stack web application enabling users to create and edit pixel art on a customizable grid-based canvas
- Integrated AI-powered prompt-based pixel art generation using the OpenAI API via a Vercel serverless backend
- Implemented a color wheel, multiple drawing modes (click and hover), and export canvas as PNG functionality

Image Processing | Python, NumPy, Matplotlib, Scikit-learn, OpenCV, [Github](#)

Jan. 2025 – Mar. 2025

- Detected edges of an image using Sobel filtering to identify intensity changes in the horizontal & vertical directions
- Reduced salt & pepper noise by applying median filtering, taking the median of a 3x3 kernel for each pixel
- Compressed image to k unique colors using K-means clustering algorithm to reduce image file size

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, OCaml, HTML, CSS, SQL

Tools, Libraries, Frameworks: Git, React, Vue, NumPy, OpenCV, Matplotlib, AutoGen